
CHARACTER SUMS
IN
NUMBER THEORY

MA262 ESSAY

APRIL 2026

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1 Introduction

1.1 Motivations

Number theorists have long been interested in finding integer solutions to systems of polynomial equations, in particular those with more variables than equations. A famous example is the Mordell ¹ Equation:

$$\text{Find } (x, y) \in \mathbb{Z}^2 \text{ such that } y^2 = x^3 + k, \ k \in \mathbb{Z} \text{ fixed.} \quad (1)$$

A common technique used is that of reducing the coefficients modulo p , and by searching for solutions in $\mathbb{F}_p^2$. Using this exact method, we can usually extract arithmetic information. Here is an example.

Theorem 1.1. [1, p. 191] *Equation (1) has no solutions if $k = (4n - 1)^3 - 4m^2$, where $m, n \in \mathbb{Z}$ such that no prime $p \equiv 3 \pmod{4}$ divides m .*

However, it turns out that there is more to equations in $\mathbb{F}_p$ than first meets the eye. They have an interesting structure in and of themselves. In fact, one of the simplest equations that one can think of in $\mathbb{F}_p$, $x^2 \equiv a \pmod{p}$ leads to the law of quadratic reciprocity, which gives us reason to believe that there is some connection between primes.

This motivates the study of polynomial equations mod p more generally. The correct tool for this turns out to be using character sums, the theory of which is developed in §2.

In §3.1, this theory is applied to the study of the equation $x^3 + y^3 = 1 \pmod{p}$, and in §3.2 to the equation $y^2 = x^3 + 1 \pmod{p}$. In the end, we develop a bridge between finding the number of solutions to said equations, and an identity for primes congruent to 1 modulo 3.

1.2 n 'th Power Residues

We first begin with the simplest case of polynomial equations.

Definition 1.1. [4, p. 45] Given a prime $p \in \mathbb{Z}_{\geq 0}$, $n \in \mathbb{Z}_{\geq 0}$ we say that $a \in \mathbb{Z}$ is an n 'th power residue mod p if there is a solution x to

$$x^n \equiv a \pmod{p}$$

Recall the following result:

¹Named for American-born British mathematician Louis Mordell (28th January 1882 - 12th March 1972).

Theorem 1.2. [5, p. 132] *Any finite subgroup of the multiplicative group of a field is cyclic.*

Theorem 1.2 implies that $\mathbb{F}_p^\times$ is cyclic. As, $\mathbb{F}_p^\times = \mathbb{F}_p \setminus \{0\}$, it follows that $\mathbb{F}_p^\times \cong C_{p-1}$, the cyclic group of order $p-1$. We now ready to resolve when an integer is an n 'th power residue mod p .

Proposition 1.1. [4, p. 45] *An integer a is an n 'th power residue mod p if and only if*

$$a^{\frac{p-1}{d}} \equiv 1 \pmod{p}, \quad \text{where } d = \gcd(n, p-1)$$

For the proof, we require the following lemma.

Lemma 1.3. [4, p. 32] *Let $a, b, m \in \mathbb{Z}_{\geq 0}$. Then the congruence $ax \equiv b \pmod{m}$ has a solution if and only if $d = \gcd(a, m)$ divides b .*

Proof. Suppose x_0 is a solution. Then $ax_0 - b = km$ for some $k \in \mathbb{Z}$. Therefore $ax_0 - km = b$, and as d divides the left hand side, it must divide b .

Suppose $d \mid b$. Then there exists $c \in \mathbb{Z}$ with $cd = b$. By the Euclidean algorithm, there exist $x_0, y_0 \in \mathbb{Z}$ such that $ax_0 + my_0 = d$. Multiplying both sides by c gives

$$ax_0c + my_0c = b$$

Reducing modulo m implies that $x = x_0c$ solves $ax \equiv b \pmod{m}$. $\square$

Proof Of Proposition 1.1. Let g be a generator of $\mathbb{F}_p^\times$. Therefore, there exist $b, c \in \mathbb{Z}$ such that $x = g^b, a = g^c$. The congruence thus becomes

$$g^{bn} \equiv g^c \pmod{p}$$

This holds if and only if $bn \equiv c \pmod{p-1}$, using the structure of $\mathbb{F}_p^\times \cong C_{p-1}$. By the lemma, this has a solution iff $d \mid c$. Then $a^{p-1/d} \equiv g^{c(p-1)/d} \equiv 1 \pmod{p}$ as c/d is an integer.

Now suppose $a^{p-1/d} \equiv 1 \pmod{p}$. Then $g^{c(p-1)/d} \equiv 1 \pmod{p}$, so $(p-1) \mid c(p-1)/d$, hence $d \mid c$. $\square$

Remark. The above proposition recovers Euler's Criterion for quadratic residues [1, p. 180]. This says that for an odd prime p , $a \in \mathbb{Z}$, $x^2 \equiv a \pmod{p}$ has a solution if and only if $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$.

1.3 The Eisenstein Integers

Definition 1.2. Let $\omega = e^{2\pi i/3} = \frac{-1+i\sqrt{3}}{2}$, a third root of unity. The *Eisenstein Integers*, $\mathbb{Z}[\omega]$, are the set

$$\mathbb{Z}[\omega] = \{a + b\omega \mid a, b \in \mathbb{Z}\} \subset \mathbb{C}$$

With addition and multiplication inherited from $\mathbb{C}$, this becomes a commutative ring, and as it is a subset of $\mathbb{C}$, it has no zero-divisors, i.e. it is an integral domain. Much like for $\mathbb{Z}[i]$, we can define a norm on $\mathbb{Z}[\omega]$ by

$$N : \mathbb{Z}[\omega] \rightarrow \mathbb{Z} : N(a + b\omega) = a^2 - ab + b^2 = (a + b\omega)\overline{(a + b\omega)}$$

N is multiplicative, i.e. $N(\alpha\beta) = N(\alpha)N(\beta)$. Under this norm, $\mathbb{Z}[\omega]$ becomes a euclidean domain [3, p.75].

We also need to extend our notion of congruence. Let $\alpha, \beta, \pi \in \mathbb{Z}[\omega]$. As in $\mathbb{Z}$, we say that $\alpha \mid \beta$ if there exists $\gamma \in \mathbb{Z}[\omega]$ such that $\beta = \alpha\gamma$. We say that $\alpha \equiv \beta \pmod{\pi}$ if $\pi \mid \alpha - \beta$, in other words, there exists $\gamma \in \mathbb{Z}[\omega]$ such that $\gamma \cdot \pi = \alpha - \beta$.

2 Character Sums

2.1 Characters

Definition 2.1. [4, p. 88] A *character* χ on $\mathbb{F}_p$ is a group homomorphism

$$\chi : \mathbb{F}_p^\times \rightarrow \mathbb{C}^\times$$

In other words, a character is a map χ such that $\chi(ab) = \chi(a)\chi(b)$ for all $a, b \in \mathbb{F}_p^\times$. As an example, one can consider the trivial character ε , defined by $\varepsilon(a) = 1$ for all $a \in \mathbb{F}_p$. Moreover, the Legendre symbol, $\rho(a) = \left(\frac{a}{p}\right)$ is also a character.² From now on, ρ will always refer to the Legendre Symbol, when the prime p is understood.

Remark. It is often useful to extend characters (as functions) to all of $\mathbb{F}_p$ by defining $\chi(0) = 0$ if $\chi \neq \varepsilon$, and $\varepsilon(0) = 1$. From now on, we consider this to be the case for all characters that we work with.

²See Appendix §4.1

Proposition 2.1. [4, p. 88] *Let χ be a character on $\mathbb{F}_p$, $a \in \mathbb{F}_p^\times$. Then*

- (a) $\chi(1) = 1$
- (b) $\chi(a)$ is a $p - 1$ 'st root of unity, i.e. $\chi(a)^{p-1} = 1$
- (c) $\chi(a^{-1}) = \chi(a)^{-1} = \overline{\chi(a)}$

Proof. For (a), notice that $\chi(1) = \chi(1 \cdot 1) = \chi(1)\chi(1)$. Hence $\chi(1)$ is the multiplicative identity in $\mathbb{C}^\times$, i.e. $\chi(1) = 1$. For (b), we have that $\chi(a)^{p-1} = \chi(a^{p-1}) = \chi(1) = 1$. For (c), the first equality follows from the fact that $1 = \chi(a^{-1}a) = \chi(a^{-1})\chi(a)$. The second equality follows because roots of unity are complex numbers of norm 1. $\square$

We define $\bar{\chi}$ to be the character $\bar{\chi}(a) := \overline{\chi(a)}$. Moreover, it is clear that given two characters χ_1, χ_2 , then $\chi_1\chi_2(a) = \chi_1(a)\chi_2(a)$ is also a character. Part (c) of Proposition 2.1 tells us that, under this multiplication, characters are invertible, so the set of characters forms a group.

Definition 2.2. [6, p. 34] The set of all characters on $\mathbb{F}_p$ under the multiplication described above is called the *dual group* of $\mathbb{F}_p$. It is denoted $\widehat{\mathbb{F}_p}$.

Definition 2.3. Let $\chi \in \widehat{\mathbb{F}_p}$ be a character. We say that χ has order d if it has order d as an element of $\widehat{\mathbb{F}_p}$.

For instance, $\rho \in \widehat{\mathbb{F}_p}$ is of order two, as it is not the identity, and $\rho^2 = \varepsilon$.

Theorem 2.1. [7, p. 38] $\widehat{\mathbb{F}_p}$ is isomorphic to $\mathbb{F}_p^\times$ as a group.

Proof. We know that $\mathbb{F}_p^\times \cong C_{p-1}$, so it suffices to show that $\widehat{\mathbb{F}_p}$ is cyclic of order $p - 1$. First, let g be a generator for $\mathbb{F}_p^\times$. A character χ is uniquely determined by $\chi(g)$. Moreover, we know that $\chi(g)$ is a $p - 1$ 'st root of unity, and there are $p - 1$ of these. It follows that there are at most $p - 1$ characters on $\mathbb{F}_p$, i.e. $|\widehat{\mathbb{F}_p}| \leq p - 1$.

To show that $\widehat{\mathbb{F}_p} \cong C_{p-1}$, it suffices to show that there is a character of order $p - 1$. Consider $\lambda : \mathbb{F}_p^\times \mapsto \mathbb{C}^\times$ given by $\lambda(g^l) = e^{2\pi il/p-1}$, $l = 1, \dots, p - 1$. It is rote verification that λ is a character.

It remains to show that λ has order $p - 1$. Suppose λ has order n . Then n is the smallest positive integer such that $\lambda^n = \varepsilon$. Then $e^{2\pi in/p-1} = 1$ for

$l = 1, \dots, p-1$. It follows that $p-1 \mid ln$. In particular, $p-1 \mid n$. As $p-1$ satisfies $\lambda^{p-1} = \varepsilon$, it follows that $n = p-1$. Thus $\widehat{\mathbb{F}_p} \cong C_{p-1}$. $\square$

Remark. Notice that in the above proof, the important fact was that $\mathbb{F}_p^\times$ is cyclic, not that it is of order $p-1$. In fact, we can define characters on C_n , the cyclic group of order n , and we have that $C_n \cong \widehat{C_n}$. For details, see the book of Schmidt [7, p. 38–40].

Proposition 2.2. [4, p. 89] *Let $\chi \in \widehat{\mathbb{F}_p}$.*

- (a) *If $\chi = \varepsilon$, then $\sum_{a \in \mathbb{F}_p} \chi(a) = p$.*
- (b) *If χ is nontrivial, then $\sum_{a \in \mathbb{F}_p} \chi(a) = 0$.*

Proof. ³ For (a), we observe that $\varepsilon(a) = 1$ for all $a \in \mathbb{F}_p$, and there are p elements in the sum.

For (b), observe that we can interpret this sum as a geometric series. Let g be a generator for $\mathbb{F}_p^\times$. Then

$$\sum_{a \in \mathbb{F}_p} \chi(a) = \chi(0) + \sum_{i=1}^{p-1} \chi(g^i) = \sum_{i=1}^{p-1} \chi^i(g) = \frac{\chi(g)(\chi^{p-1}(g) - 1)}{\chi(g) - 1} = 0$$

Observing that $\chi(g) \neq 1$ as χ is nontrivial. $\square$

Remark. Specialising the above to the Legendre symbol, we have $\sum_a \rho(a) = 0$. This implies that there are as many residues as there are non-residues mod p .

Proposition 2.3. [4, p. 89] *Let $1 \neq a \in \mathbb{F}_p^\times$. Then $\sum_{\chi \in \widehat{\mathbb{F}_p}} \chi(a) = 0$.*

Proof. Once again, we can interpret the sum as a geometric series. Let $a = g^l$, g a generator of $\mathbb{F}_p^\times$, and recall that $\widehat{\mathbb{F}_p}$ is generated by the character λ , given by $\lambda(g) = e^{2\pi i/p-1}$. Then, we have that

$$\sum_{\chi \in \widehat{\mathbb{F}_p}} \chi(a) = \sum_{k=1}^{p-1} \lambda^k(g^l) = \sum_{k=1}^{p-1} e^{2\pi i k l / p-1} = \frac{e^{2\pi i l} (e^{2\pi i k l} - 1)}{e^{2\pi i l / p-1} - 1} = 0$$

$\square$

³The proofs of Propositions 2.2 and 2.3 are my own.

We now get to the first remarkable result: there is a very direct connection between characters and counting the number of solutions.

Theorem 2.2. [4, p. 89] *Denote by $N_p(x^n = a)$ the number of solutions to $x^n = a$ in $\mathbb{F}_p$. Let $d = \gcd(n, p - 1)$. Then:*

$$\sum_{\chi^d = \varepsilon} \chi(a) = N_p(x^n = a)$$

Where the sum runs over all characters of order dividing d .

Before the proof we require a Lemma.

Lemma 2.3. *Suppose $d \mid p - 1$. Then there are d characters whose order divides d in $\widehat{\mathbb{F}_p}$. Moreover, they form a cyclic subgroup of order d .*

Proof. Let g be a generator of $\mathbb{F}_p$. A character of order d must map g to an d 'th root of unity, so there are at most d such characters. Set $\chi_1(g) = e^{2\pi i/d}$. One can verify that this is a character on $\mathbb{F}_p$, and that $\varepsilon, \chi_1, \chi_1^2, \dots, \chi_1^{d-1}$ are distinct characters. (This is very similar to the proof of Theorem 2.1). $\square$

Remark. Lemma 2.3 implies that the Legendre symbol ρ is the unique character of order 2.

Proof of Theorem 2.2. First consider the case $a = 0$. Then $\chi(0) = 0$ for nontrivial characters, and $\varepsilon(0) = 1$. Hence the sum evaluates to 1 and there is one solution to $x^n = 0$.

Suppose now that $x^n = a$ is solvable. We know that there are d solutions, and that $d \mid c$, where $a = g^c$ (see the proof of Proposition 1.1). Set $c = dk$. Then,

$$\sum_{\chi^d = \varepsilon} \chi(a) = \sum_{\chi^d = \varepsilon} \chi(g)^c = \sum_{\chi^d = \varepsilon} \chi(g)^{dk} = d$$

Each term of the sum is equal to 1 as each is a character of order dividing d , and there are d summands by Lemma 2.3.

Lastly, suppose that $x^n = a$ is not solvable. Then $d \nmid c$, so $\chi_1(g^c) = e^{2\pi ic/d} \neq 1$. Mimicking the proof of Proposition 2.3, we can conclude that the sum is 0. $\square$

If p is an odd prime, then a character of order 2 exists, which must be the Legendre symbol by the Remark. It follows that $N_p(x^2 = a) = 1 + \rho(a)$. Furthermore, if $p \equiv 1 \pmod{3}$, then a character of order 3 exists, call it χ . Then $N_p(x^3 = a) = 1 + \chi(a) + \chi^2(a)$.

2.2 Gauss Sums

Definition 2.4. [4, p. 91] Let $a \in \mathbb{F}_p$, χ a character on $\mathbb{F}_p$, $\zeta = e^{2\pi i/p}$. Set $g_a(\chi) = \sum_{t \in \mathbb{F}_p} \chi(t) \zeta^{at}$. We call $g_a(\chi)$ the *Gauss Sum belonging to χ* .

Remark. One may wonder why we are interested in Gauss Sums. The key property is that $|g_a(\chi)|^2 = p$ ⁴, Gauss based his final proof of the law of Quadratic Reciprocity on this fact⁵. He also recognised that this is the appropriate tool for studying higher reciprocity laws. Curiously, Gauss only ever studied the Gauss Sum in the case when χ is a character of order 2, and the above definition was first given by Dirichlet. [2, p. 1]

Lemma 2.4. [4, p. 71] Let ζ be as above. Then $\sum_{t=0}^{p-1} \zeta^t = 0$.

Proof. ⁶ We know that ζ and 1 are roots of $x^p - 1$. Dividing by $x - 1$ gives $x^p - 1 = (x - 1)(x^{p-1} + x^{p-2} + \cdots + x + 1)$. Evaluating at ζ gives us that $1 + \zeta + \cdots + \zeta^{p-1} = 0$, which is exactly the sum above. $\square$

Proposition 2.4. [4, p. 91] Let $a \in \mathbb{F}_p$, $\chi \in \widehat{\mathbb{F}_p}$. Then $\chi(a)g_a(\chi) = g_1(\chi)$.

Proof. This is simply a computation.

$$\chi(a)g_a(\chi) = \chi(a) \sum_{t \in \mathbb{F}_p} \chi(t) \zeta^{at} = \sum_{t \in \mathbb{F}_p} \chi(at) \zeta^{at} = g_1(\chi)$$

Using the fact that as t runs over $\mathbb{F}_p$, so does at . $\square$

The last three results will find their use in the next (and very important) proposition.

Proposition 2.5. [7, p. 47] Let $a \in \mathbb{F}_p$, $\chi \in \widehat{\mathbb{F}_p}$.

- (a) If $\chi = \varepsilon$, then $g_a(\chi) = 0$.
- (b) If χ is nontrivial, then $|g_a(\chi)| = \sqrt{p}$.

⁴Proposition 2.5

⁵A modern exposition can be found in the book of Apostol. [1, §9.9]

⁶This proof is my own.

Proof. For (a), we observe that $g_a(\varepsilon) = \sum_t \varepsilon(at) \zeta^{at} = \sum_t \zeta^{at} = 0$ by Lemma 2.4.

For (b), we begin by observing that it suffices to prove the proposition in the case $a = 1$, as $|g_a(\chi)| = |\chi(a)g_1(\chi)| = |g_1(\chi)|$ by Proposition 2.1. Moreover, as $g_1(\chi) \in \mathbb{C}$, (b) is equivalent to $g_1(\chi)g_1(\chi) = p$.

$$\begin{aligned}
g_1(\chi)\overline{g_1(\chi)} &= \left(\sum_x \chi(x) \zeta^x \right) \overline{\left(\sum_y \chi(y) \zeta^y \right)} \\
&= \sum_{x,y} \chi(x) \overline{\chi(y)} \zeta^{x-y} \\
&= \sum_{t,y} \chi(ty) \overline{\chi(y)} \zeta^{ty-y} \quad \text{By setting } x = ty \\
&= \sum_t \chi(t) \sum_y \chi(y) \overline{\chi(y)} \zeta^{(t-1)y} \\
&= \sum_t \chi(t) \sum_y \zeta^{(t-1)y}
\end{aligned}$$

Now, observe that if $t \neq 1$, then $\sum_y \zeta^{(t-1)y} = 0$ by Lemma 2.4, observing that $(t-1)y$ runs over $\mathbb{F}_p$ as y does. If $t = 1$, then said sum is equal to p . Thus, we have $|g_1(\chi)|^2 = \chi(1)p$, which implies $|g_1(\chi)| = \sqrt{p}$. $\square$

Remark. We can formalise the 'change of variables' in the above sum as follows. For each y , the map $t \mapsto ty$ gives a bijection of $\mathbb{F}_p$ with itself. Therefore, as x runs over all elements of $\mathbb{F}_p$, so does ty . The same justification can be used for the map $y \mapsto (t-1)y$. For $t \neq 1$, this is a bijection. In both cases, we can justify this being a bijection simply by solving for the relevant variable.

2.3 Jacobi Sums

Definition 2.5. [4, p. 93] Let $\chi, \psi \in \widehat{\mathbb{F}_p}$. We call

$$J(\chi, \psi) = \sum_{\substack{a, b \in \mathbb{F}_p \\ a+b=1}} \chi(a) \psi(b) = \sum_{a \in \mathbb{F}_p} \chi(a) \psi(1-a)$$

the *Jacobi sum associated to χ and ψ* .

Remark. For readability, we suppress the explicit declaration that $a, b \in \mathbb{F}_p$ in the above sum.

The following proposition motivates the introduction of Jacobi Sums.

Proposition 2.6. [4, p. 95] *Let $p \equiv 1 \pmod{3}$. Then there exists a character $\chi \in \widehat{\mathbb{F}_p}$ of order 3 such that*

$$N_p(x^3 + y^3 = 1) = p + J(\chi, \chi^2) + J(\chi^2, \chi) + J(\chi, \chi) + J(\chi^2, \chi^2)$$

Proof. Firstly, we observe that

$$N_p(x^3 + y^3 = 1) = \sum_{a+b=1} N_p(x^3 = a)N_p(x^3 = b)$$

The subgroup of characters of order 3 is cyclic by Lemma 2.3. Let χ be a generator for this subgroup. Applying Theorem 2.2, the above sum becomes

$$\sum_{a+b=1} N_p(x^3 = a)N_p(x^3 = b) = \sum_{a+b=1} (1 + \chi(a) + \chi^2(a))(1 + \chi(b) + \chi(b)^2)$$

Expanding the product on the right hand side, noticing that there are p elements in the sum, and using Proposition 2.2 gives the result. $\square$

Remark. We have reduced the problem of counting solutions in $\mathbb{F}_p$ to the problem of understanding the Jacobi Sum. Notice too, that there is nothing special about the equation $x^3 + y^3 = 1$. We can do the exact same thing for other equations. For instance, assuming $p \equiv 1 \pmod{n}$, the exact same argument as above works for $x^n + y^n = 1$.

Theorem 2.5. [4, p. 93] *Let $\chi, \psi \in \widehat{\mathbb{F}_p}$ such that $\chi, \psi \neq \varepsilon$. Then we have the following*

$$(a) \ J(\varepsilon, \varepsilon) = p$$

$$(b) \ J(\varepsilon, \chi) = 0$$

$$(c) \ J(\chi, \chi^{-1}) = -\chi(-1)$$

(d) *Suppose further that the product is nontrivial: $\chi\psi \neq \varepsilon$. Then:*

$$J(\chi, \psi) = \frac{g_1(\chi)g_1(\psi)}{g_1(\chi\psi)}$$

Proof. (a) and (b) are immediate consequences of the definition.

For part (c), we use the second characterisation of the Jacobi sum associated to $\chi, \bar{\chi}$. As χ is non-trivial, $\chi(0) = 0$. Therefore, we may exclude the

elements of the sum where $a = 0, 1$.

$$J(\chi, \bar{\chi}) = \sum_{a \neq 0, 1} \chi(a) \chi(1-a) = \sum_{a \neq 0, 1} \chi\left(\frac{a}{1-a}\right)$$

Set $b = a/(1-a)$. As a runs through $\mathbb{F}_p \setminus \{0, 1\}$, b runs over $\mathbb{F}_p \setminus \{0, -1\}$, this can be seen as $a = b/(1+b)$.⁷ Then, we have that

$$J(\chi, \chi^{-1}) = \sum_{b \neq 0, -1} \chi(b) = \sum_{b \neq 0} \chi(b) - \chi(-1) = -\chi(-1)$$

The last equality follows from Proposition 2.2, noting that $\chi(0) = 0$.

For part (d), we begin by expanding $g_1(\chi)g_1(\psi)$.

$$\begin{aligned} g_1(\chi)g_1(\psi) &= \left(\sum_{x \in \mathbb{F}_p} \chi(x) \zeta^x \right) \left(\sum_{y \in \mathbb{F}_p} \psi(y) \zeta^y \right) \\ &= \sum_{x, y \in \mathbb{F}_p} \chi(x) \psi(y) \zeta^{x+y} \\ &= \sum_{t \in \mathbb{F}_p} \left(\sum_{x+y=t} \chi(x) \psi(y) \zeta^{x+y} \right) \end{aligned}$$

Split up the inner sum into two cases. First, if $t = 0$, then $x = -y$. The inner sum becomes

$$\sum_x \chi(x) \psi(-x) = \psi(-1) \sum_x \chi \psi(x) \stackrel{\text{Prop 2.2}}{=} 0$$

Make the change of variables $x = tx'$, and $y = ty'$ to get

$$\begin{aligned} \sum_{t \in \mathbb{F}_p^\times} \left(\sum_{x+y=t} \chi(x) \psi(y) \zeta^{x+y} \right) &= \sum_{t \in \mathbb{F}_p^\times} \left(\sum_{x'+y'=1} \chi(t) \psi(t) \chi(x') \chi(y') \zeta^t \right) \\ &= \sum_{t \in \mathbb{F}_p^\times} \chi \psi(t) \zeta^t \left(\sum_{x'+y'=1} \chi(x') \psi(y') \right) \\ &= J(\chi, \psi) \sum_{t \in \mathbb{F}_p^\times} \chi \psi(t) \zeta^t \end{aligned}$$

⁷See the remark after Proposition 2.5.

Observe that as $\chi\psi \neq \varepsilon$, $\chi\psi(0) = 0$. Therefore:

$$\sum_{t \in \mathbb{F}_p^\times} \chi\psi(t)\zeta^t = \sum_{t \in \mathbb{F}_p} \chi\psi(t)\zeta^t = g_1(\chi\psi)$$

Putting everything together gives us

$$g_1(\chi)g_1(\psi) = J(\chi, \psi)g_1(\chi\psi)$$

This is exactly what we wanted to show. $\square$

Remark. It is a common technique when working with double sums to introduce another variable, and rewrite the sum in terms of ' $x + y = t$ '. This technique can be seen in the proof of Proposition 4.4.

Corollary. [4, p. 94] *If χ, ψ are nontrivial characters on $\mathbb{F}_p$, then $|J(\chi, \lambda)| = \sqrt{p}$.*

Proof.

$$|J(\chi, \psi)| \stackrel{\text{Thm 2.5}}{=} \frac{|g_1(\chi)g_1(\psi)|}{|g_1(\chi\psi)|} \stackrel{\text{Prop 2.5}}{=} \frac{p}{\sqrt{p}} = \sqrt{p}$$

$\square$

We can now go further in our analysis of $N_p(x^3 + y^3 = 1)$.

Proposition 2.7. [4, p. 95] Let $p \equiv 1 \pmod{3}$. Then there exists $\chi \in \widehat{\mathbb{F}_p}$ such that

- (a) $N_p(x^3 + y^3 = 1) = p - 2 + J(\chi, \chi) + J(\chi^2, \chi^2)$
- (b) $|N_p(x^3 + y^3 = 1) - p + 2| \leq 2\sqrt{p}$

Proof. For part (a), recall the result of Proposition 2.6:

$$N_p(x^3 + y^3 = 1) = p + J(\chi, \chi^2) + J(\chi^2, \chi) + J(\chi, \chi) + J(\chi^2, \chi^2) \quad (2)$$

Note that $J(\chi, \chi^2) = J(\chi, \chi^{-1})$. Therefore, we can use Theorem 2.5 (c), to see that $J(\chi, \chi^{-1}) = -\chi(-1)$. Now, $\chi(-1) = \chi((-1)^3) = \chi^3(-1) = 1$ as χ is of order 3. (2) therefore becomes

$$N_p(x^3 + y^3 = 1) = p + 2J(\chi, \chi^{-1}) + J(\chi, \chi) + J(\chi^2, \chi^2) \quad (3)$$

$$= p - 2 + J(\chi, \chi) + J(\chi^2, \chi^2) \quad (4)$$

For (b), rearrange (4) and take absolute values to get

$$\begin{aligned}
|N_p(x^3 + y^3 = 1) - p + 2| &= |J(\chi, \chi) + J(\chi^2 + \chi^2)| \\
&\leq |J(\chi, \chi)| + |J(\chi^2, \chi^2)| \\
&= 2\sqrt{p}
\end{aligned}
\tag{Thm 2.5, (d)}$$

□

What we are saying is that the number of solutions to $x^3 + y^3 = 1$ in $\mathbb{F}_p$ is approximately $p - 2$ with an 'error term' of $2\sqrt{p}$. This analysis can be extended to equations of the form $a_1x^{r_1} + \dots + a_nx^{r_n} = b$. This was done in a paper of Weil [8], which is worth reading for its mathematical significance - here Weil formulated what became known as the Weil Conjectures, which have led to great progress in Number Theory.

3 Applications

3.1 The Equation $x^3 + y^3 = 1$ in $\mathbb{F}_p$

Finishing the analysis of the equation $x^3 + y^3 = 1$ (p) will yield some beautiful number-theoretic results. We begin with the following observation. Observe that if χ has order 3, then $\chi(a)$ is a 3rd root of unity for all $a \in \mathbb{F}_p^\times$. Therefore, $J(\chi, \chi)$ is a sum of elements of $\mathbb{Z}[\omega]$ - the eisenstein integers (§1.3). We therefore get the following result.

Proposition 3.1. [4, p. 95] *Let $p \equiv 1 \pmod{3}$, $\chi \in \widehat{\mathbb{F}_p}$ is a character of order 3. Then, there exist $a, b \in \mathbb{Z}$ such that $4p = (2a - b)^2 + 3b^2$.*

Proof. As $J(\chi, \chi) \in \mathbb{Z}[\omega]$, there exist $a, b \in \mathbb{Z}$ such that $a + b\omega = J(\chi, \chi)$. By the corollary to Theorem 2.5, we have $|J(\chi, \chi)| = \sqrt{p}$. Therefore,

$$p = |J(\chi, \chi)|^2 = (a + b\omega)(a + b\bar{\omega}) = a^2 - ab + b^2$$

Where the last equality follows from the fact that $\omega = \frac{-1+i\sqrt{3}}{2}$. It follows that $4p = 4a^2 - 4ab + 4b^2 = (2a - b)^2 + 3b^2$. □

It turns out that we can say a lot more about the a, b that we have found above. Moreover, we can link this back to counting solutions. First, we require a few more preliminary results.

Proposition 3.2. [4, p. 96] *Suppose that $p \equiv 1 \pmod{3}$, and that $\chi \in \widehat{\mathbb{F}_p}$ is a character of order 3. Then*

- (a) $g(\chi)^3 = pJ(\chi, \chi)$
(b) $J(\chi^2, \chi^2) = J(\bar{\chi}, \bar{\chi}) = \overline{J(\chi, \chi)}$

For the proof we require a small lemma.

Lemma 3.1. [4, p. 91] *Let $\chi \in \widehat{\mathbb{F}_p}$. Then $\overline{g_1(\chi)} = \chi(-1)g_1(\bar{\chi})$*

Proof.

$$\begin{aligned} \overline{g_1(\chi)} &= \sum_{t \in \mathbb{F}_p} \overline{\chi(t)\zeta^t} = \sum_{t \in \mathbb{F}_p} \bar{\chi}(t)\zeta^{-t} = \bar{\chi}(-1) \sum_{t \in \mathbb{F}_p} \bar{\chi}(-t)\zeta^{-t} \\ &= \chi(-1)g_1(\bar{\chi}) \end{aligned}$$

The last equality follows as $\chi(-1) = \pm 1$, hence $\bar{\chi}(-1) = \chi(-1)$. $\square$

Proof of Proposition 3.2. For (a). Recall that $\chi(-1) = 1$ when χ is a cubic character. By Theorem 2.5 (d), $J(\chi, \chi)g_1(\chi^2) = g_1(\chi)^2$. Lemma 3.1 implies $g_1(\chi^2) = g_1(\chi^{-1}) = \overline{g_1(\chi)}$. Multiplying both sides of $J(\chi, \chi)g_1(\chi^2) = g_1(\chi)^2$ by $g_1(\chi)$ gives:

$$J(\chi, \chi)\overline{g_1(\chi)}g_1(\chi) = g_1(\chi)^3$$

By Proposition 2.5, the left hand side becomes $pJ(\chi, \chi)$, and (a) follows.

For (b), the first equality is immediate from the fact that $\chi^2 = \bar{\chi}$, as χ has order 3. For the second equality, apply Theorem 2.5 (d):

$$J(\bar{\chi}, \bar{\chi}) = \frac{g_1(\bar{\chi})g_1(\bar{\chi})}{g_1(\bar{\chi}^2)} \stackrel{\text{Lemma 3.1}}{=} \frac{\overline{g_1(\chi)g_1(\chi)}}{g_1(\chi^2)} = \overline{J(\chi, \chi)}$$

$\square$

Proposition 3.3. [4, p. 96] *Let $p \equiv 1 \pmod{3}$, $\chi \in \widehat{\mathbb{F}_p}$ of order 3. Set $J(\chi, \chi) = a + b\omega$ for some $a, b \in \mathbb{Z}$. Then:*

- (a) $a \equiv -1 \pmod{3}$
(b) $b \equiv 0 \pmod{3}$

Proof. Using congruences in $\mathbb{Z}[\omega]$, we calculate:

$$a + b\omega \equiv J(\chi, \chi) \stackrel{\text{Prop 3.1}}{\equiv} g(\chi)^3 \equiv \sum_{t \in \mathbb{F}_p} \chi^3(t)\zeta^{3t} \equiv \sum_{t \in \mathbb{F}_p^\times} \zeta^{3t} \pmod{3}$$

Where the second congruence follows from the binomial coefficients $\binom{3}{k}$ being divisible by 3 and the last congruence from the fact that $\chi^3(t) = 1$ if t is nonzero and is 0 otherwise.

As t runs through $\mathbb{F}_p^\times$, so does $3t$. Lemma 2.4 implies that $\sum \zeta^{3t} + \zeta^0 = 0$, so $a + b\omega \equiv -1$ modulo 3.

Considering $\overline{J(\chi, \chi)} = a + b\bar{\omega}$, and doing the same thing as above we see that $a + b\bar{\omega} \equiv -1$ (3).

Then, $a + b\omega - (a + b\bar{\omega}) = b(\omega - \bar{\omega}) = b(\sqrt{-3}) \equiv 0$ (3). Therefore $-3b^2 \equiv 0$ (9), and so $3 \mid b$. As $a + b\omega \equiv -1$ (3), then $a \equiv -1$ (3). $\square$

Immediately, as $b \equiv 0$ (3), we can set $B = b/3$. Substituting into the result of Proposition 3.1, we get $4p = (2a - b)^2 + 27B^2$.

Recalling Proposition 2.7, we have reduced $N_p(x^3 + y^3 = 1)$ to

$$N_p(x^3 + y^3 = 1) = p - 2 + 2\operatorname{Re}J(\chi, \chi)$$

As $J(\chi, \chi) = a + b\omega$, we can explicitly calculate the real part.

$$J(\chi, \chi) = a + b\omega = a - \frac{b}{2} + i\frac{-b\sqrt{3}}{2}$$

So, $2\operatorname{Re}J(\chi, \chi) = 2a - b$. This is exactly the same as in our identity for $4p$! Setting $A = 2a - b$, the above proposition also implies that $A \equiv 1$ (3). We summarise this in the following crowning theorem of the essay.

Theorem 3.2. [4, p. 97] *Let $p \equiv 1$ (3). Then there exist integers A, B such that $A \equiv 1$ (3) and*

$$(a) \quad 4p = A^2 + 27B^2$$

$$(b) \quad N_p(x^3 + y^3 = 1) = p - 2 + A$$

Thus, we have established some connection between the identities for primes and counting solutions to equations over finite fields. A few examples of the above result:⁸

- $4 \cdot 13 = (-5)^2 + 27 \cdot 1^2$ hence $N_{13}(x^3 + y^3 = 1) = 6$
- $4 \cdot 37 = (-11)^2 + 27 \cdot 1^2$ hence $N_{37}(x^3 + y^3 = 1) = 24$
- $4 \cdot 97 = 19^2 + 27 \cdot 1^2$ hence $N_{97}(x^3 + y^3 = 1) = 114$

⁸[4, §8, Exercise 11]

3.2 Mordell's Equation over $\mathbb{F}_p$

We now return to Mordell's Equation. Inspired by what we have done with $x^3 + y^3 = 1$, we see if there is anything similar we can find out from $y^2 = x^3 + k$ (p). In this section, we assume that $p \equiv 1 \pmod{6}$.

Theorem 3.3. [4, §8, Exercise 15] *Suppose that $p \equiv 1 \pmod{6}$. Let χ, ρ be characters of orders 3, 2 respectively. Then $N_p(y^2 = x^3 + k) = p + \pi + \bar{\pi}$, where $\pi = \chi\rho(k)J(\chi, \rho)$.*

Proof. First, we observe that $N_p(x^3 = b) = N_p(x^3 = -b)$ as $a^3 = b$ if and only if $(-a)^3 = -b$. Rearranging then gives $N_p(x^3 = b) = N_p(-x^3 = b)$. As $p \equiv 1 \pmod{6}$, it follows that $p \equiv 1 \pmod{2}$ and $p \equiv 1 \pmod{3}$. Therefore, we may apply Theorem 2.2 to count solutions as before.

$$\begin{aligned} N_p(y^2 = x^3 + k) &= N_p(y^2 - x^3 = k) = \sum_{a+b=k} N_p(y^2 = a)N_p(-x^3 = b) \\ &= \sum_{a+b=k} N_p(y^2 = a)N_p(x^3 = b) \\ &= \sum_{a+b=k} (1 + \rho(a))(1 + \chi(b) + \chi^2(b)) \end{aligned}$$

Now, make the substitution $a = ka'$, $b = kb'$. After simplifying, we get:

$$N_p(y^2 = x^3 + k) = p + \chi\rho(k)J(\chi, \rho) + \chi^2\rho(k)J(\chi^2, \rho)$$

It is not difficult to see that $\overline{\chi\rho(k)J(\chi, \rho)} = \chi^2\rho(k)J(\chi^2, \rho)$. Therefore $N_p(y^2 = x^3 + k) = p + \pi + \bar{\pi}$, as required. $\square$

We can do more, but for that, we require some slightly more technical results. These are presented in Appendix 4.2. The final result that we will need is:

Proposition 3.4. [4, §8, Exercise 6] *Let χ, ρ be as above. Then $J(\chi, \chi) = \chi(2^{-2})J(\chi, \rho)$.*

Corollary. *Suppose that $\chi(2) = 1$. Then $N_p(y^2 = x^3 + 1) = p + A$, where $A \equiv 1 \pmod{3}$ and $4p = A^2 + 27B^2$.*

Proof. We know that $N_p(y^2 = x^3 + 1) = p + \pi + \bar{\pi}$, where $\pi = \chi\rho(1)J(\chi, \rho)$. We know that $\psi(1) = 1$ for any character, so the above simplifies to $p +$

$J(\chi, \rho) + \overline{J(\chi, \rho)}$. Applying Proposition 3.4 above, we get

$$\begin{aligned}
N_p(y^2 = x^3 + 1) &= p + \chi(2)^2 J(\chi, \chi) + \overline{\chi(2)^2 J(\chi, \chi)} \\
&= p + J(\chi, \chi) + \overline{J(\chi, \chi)} \\
&= p + 2\operatorname{Re} J(\chi, \chi) \\
&= p + A
\end{aligned}$$

□

We apply this result with the prime $p = 31$. We know that a cubic character χ is given by $\chi(g) = e^{2\pi i/3} = \omega$, where g is a generator of $\mathbb{F}_{31}^\times$. We may take $g = 3$ as our generator. Then $\chi(2) = \chi(3^{24}) = \omega^{24} = 1$.

Now, we write $4(31) = 4^2 + 27(2)^2$, noticing that $4 \equiv 1 \pmod{3}$. The Corollary implies that $N_{31}(y^2 = x^3 + 1) = 35$. A quick computer search verifies this is the case.⁹

Table 1: Solutions (x, y) to $y^2 = x^3 + 1 \pmod{31}$

(0, 1)	(0, 30)	(1, 8)	(1, 23)	(2, 3)
(2, 28)	(3, 11)	(3, 20)	(5, 8)	(5, 23)
(6, 0)	(10, 3)	(10, 28)	(13, 11)	(13, 20)
(15, 11)	(15, 20)	(16, 6)	(16, 25)	(17, 4)
(17, 27)	(18, 6)	(18, 25)	(19, 3)	(19, 28)
(22, 4)	(22, 27)	(23, 4)	(23, 27)	(25, 8)
(25, 23)	(26, 0)	(28, 6)	(28, 25)	(30, 0)

⁹For the Code, see Appendix 4.3

4 Appendix

4.1 Quadratic Reciprocity

Definition 4.1. Let $a \in \mathbb{Z}$. The *Legendre Symbol* is defined to be $\left(\frac{a}{p}\right) := a^{\frac{p-1}{2}} (p)$.

Note that if $p \mid a$, then $\left(\frac{a}{p}\right) = 0$. Moreover, as $a^{p-1} \equiv 1 (p)$, we can factorise as follows:

$$(a^{\frac{p-1}{2}} - 1)(a^{\frac{p-1}{2}} + 1) \equiv 0 (p)$$

So $a^{\frac{p-1}{2}} \equiv \pm 1 (p)$. By Proposition 1.1, this is equal to 1 iff a is a quadratic residue. Therefore, the legendre symbol is equal to -1 if and only if a is a non-residue mod p . In this sense, the legendre symbol detects residues. Moreover, the following proposition says that the Legendre symbol is a character.

Proposition 4.1. Let $a, b \in \mathbb{Z}$. Then the following hold.

$$(a) \left(\frac{a}{p}\right) \left(\frac{b}{p}\right) = \left(\frac{ab}{p}\right)$$

$$(b) \text{ If } a \equiv b (p), \text{ then } \left(\frac{a}{p}\right) = \left(\frac{b}{p}\right)$$

The proof follows from the definition.

4.2 Technical Results

This section is dedicated to my solutions to some of the exercises in [4, Chapter 8], building up to Proposition 3.4. These would have been in the main section of the essay, however they were slightly too lengthy.

Proposition 4.2. [4, §8, Exercise 3] Let p be an odd prime, χ a non-trivial character, ρ the character of order 2. Then $\sum_t \chi(1 - t^2) = J(\chi, \rho)$.

Proof. $J(\chi, \rho) = \sum_{a+b=1} \chi(a)\rho(b) = \sum_a \chi(a)\rho(1-a)$ Now, we use that $N_p(x^2 = 1 - a) = 1 + \rho(1 - a)$. Making this substitution, we get

$$J(\chi, \rho) = \sum_a \chi(a)N_p(x^2 = 1 - a) - \chi(a) \stackrel{\text{Prop 2.2}}{=} \sum_a \chi(a)N_p(x^2 = 1 - a)$$

If $1 - a = 0$, then it is counted once in the sum, as $N_p(x^2 = 1 - a) = 1$. This leaves us with $p - 1$ terms in the sum. If $1 - a$ is a quadratic residue, then it

is counted twice, as then $N_p(x^2 = 1 - a) = 2$. If it is a non-residue, then it isn't counted. As there are $p - 1/2$ residues, it follows that:

$$\sum_a \chi(a) N_p(x^2 = 1 - a) = \sum_t \chi(1 - t^2)$$

□

Proposition 4.3. [4, §8, Exercise 4] *Let p be an odd prime, χ a nontrivial character, ρ the character of order 2. Suppose $k \in \mathbb{F}_p^\times$. Then $\sum_t \chi(t(k-t)) = \chi(2^2/k^2)J(\chi, \rho)$.*

Proof. This follows from Proposition 4.3 in the following way:

$$\begin{aligned} J(\chi, \rho) &= \sum_t \chi \left(1 - \left(\frac{k-2t}{k} \right)^2 \right) \\ &= \sum_t \chi \left(1 - \frac{k^2 - 4kt + 4t^2}{k^2} \right) \\ &= \sum_t \chi \left(\frac{4tk - 4t^2}{k^2} \right) \\ &= \frac{\chi(2^2)}{\chi(k^2)} \sum_t \chi(t(k-t)) \end{aligned}$$

Noting that as t runs through $\mathbb{F}_p$, so does $(k-2t)/k$.

□

Proposition 4.4. [4, §8, Exercise 6] *Suppose that $\chi^2 \neq \varepsilon$. Then $g_1(\chi)^2 = \chi(2)^{-2} J(\chi, \rho) g_1(\chi^2)$.*

Proof. This is simply a calculation.

$$\begin{aligned}
g_1(\chi)^2 &= \left(\sum_t \chi(t) \zeta^t \right) \left(\sum_s \chi(s) \zeta^s \right) = \sum_{t,s} \chi(t) \chi(s) \zeta^{t+s} \\
&= \sum_k \left(\sum_{s+t=k} \chi(s) \chi(t) \right) \zeta^k \\
&= \sum_k \left(\sum_{s+t=k} \chi(k-t) \chi(t) \right) \zeta^k \\
&= \sum_k \left(\sum_t \chi(t(k-t)) \right) \zeta^k
\end{aligned}$$

We are now in a position to use Proposition 4.3. However, we need to be careful, as that proposition is only valid for $k \neq 0$. So, we split the sum up accordingly.

$$\begin{aligned}
\sum_k \left(\sum_t \chi(t(k-t)) \right) \zeta^k &= \sum_{k \neq 0} \left(\sum_t \chi(t(k-t)) \right) \zeta^k - \sum_t \chi^2(t) \\
&= \sum_{k \neq 0} J(\chi, \rho) \chi(k^2) \chi(2)^{-2} \zeta^k \\
&= J(\chi, \rho) \chi(2)^{-2} \sum_{k \neq 0} \chi^2(k) \zeta^k
\end{aligned}$$

Here, we notice that $\sum_{k=0} \chi^2(k) \zeta^k = \sum_k \chi^2(k) \zeta^k$ as χ^2 is a nontrivial character by hypothesis, and we have defined $\psi(0) = 0$ for nontrivial characters ψ . This is exactly the definition of $g_1(\chi^2)$, so we have that

$$g_1(\chi)^2 = J(\chi, \rho) \chi(2)^{-2} g_1(\chi^2)$$

as required. □

Proof of Proposition 3.4. Apply Theorem 2.5, (d).

$$\begin{aligned}
 J(\chi, \chi) &= \frac{g_1(\chi)g_1(\chi)}{g_1(\chi^2)} = \frac{g_1(\chi)^2}{g_1(\chi^2)} \\
 &= \frac{J(\chi, \rho)\chi(2)^{-2}g_1(\chi^2)}{g_1(\chi^2)} && \text{By Prop 4.4} \\
 &= J(\chi, \rho)\chi(2)^{-2}
 \end{aligned}$$

□

4.3 Searching for Solutions to $y^2 = x^3 + 1$ (31).

This is the python code used for Table 1.

```

def mordell_mod_p(x,y):
    return (y**2 - x**3) % 31

sol_count =0

for x in range(31):
    for y in range(31):
        if mordell_mod_p(x,y) == 1:
            sol_count +=1
            print(f"({x},{y})")

print(sol_count)

```

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